

CFC

Control Framework for Closure

Independent AI Evaluation & Reliability Project | Technical Brief | September 2026

Problem CFC targets: an LLM produces a definite conclusion even though the evidence or current state is still insufficient, conflicting, stale, out of scope or unresolved.

What CFC does

CFC is a deterministic evaluation/control layer built around explicit state, evidence and closure rules. It separates the model's ability to generate an answer from whether the available state actually permits that answer to be closed.

- Keeps UNRESOLVED distinct from positive and negative outcomes.
- Checks validity, scope, provenance, conflict resolution and state binding before closure.
- Fails closed when required evidence is incomplete or invalid.
- Returns an auditable reason for no-closure instead of guessing missing state.

Representative failure modes

- UNRESOLVED silently becomes TRUE or FALSE.
- Conflicting records are resolved without a valid rule.
- Stale, inactive or wrong-scope evidence influences the result.
- A definitive answer is produced while required checks remain incomplete.

Current state

- Working framework with frozen baseline v1.23.
- Frozen baseline remains byte-for-byte unchanged during current review work.
- External formal state-and-closure specification is being developed to describe existing behavior.
- Internal test results are encouraging; current priority is independent review and external replication.

How it is evaluated

CFC is tested as a reliability problem, not as prompt tuning. Cases are constructed so that the correct outcome depends on the formal evidence state rather than on what answer sounds plausible.

- Frozen baselines prevent post-hoc controller changes during evaluation.
- Adversarial and edge cases cover conflict, scope, stale evidence, state transitions and resolution validity.
- Preserved model outputs can be replayed without repeated API cost.
- External review is separated from internal development testing.

Why this matters for LLM / agent evaluation

Many evaluation systems ask whether an answer is correct, useful, safe or grounded. CFC asks a narrower underlying question: was the system actually entitled to reach a definitive conclusion from the state it had?

For agentic systems, an unsupported conclusion can propagate into a later tool call, action or decision, turning a reasoning error into an operational one.

What I am looking for

I am looking to connect with people working in LLM/agent evaluation, AI reliability, model behavior, validation or AI governance who may be interested in an independent technical look at CFC.

- Technical critique of the state/closure model.
- Independent replay or replication of representative cases.
- Comparison with existing evaluation tooling and failure-mode taxonomies.
- Discussion of practical integration into evaluation or agent pipelines.

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